

TITLE OF PROJECT REPORT

SMART OPD SYSTEM

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report titled "SMART OPD SYSTEM" is the bonafide work of KAUSHAL TANNA, VYOM DESAI, DAKSH LATHIYA, PRASHIL DHINGANI, and VISHESH JAIN, who carried out the project work under my supervision. Certified further that, to the best of my knowledge, the work reported herein does not form part of any other project or research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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LIST OF ABBREVIATIONS

OPD	— Outpatient Department
API	— Application Programming Interface
SPA	— Single Page Application
REST	— Representational State Transfer
JWT	— JSON Web Token
JSON	— JavaScript Object Notation
UI/UX	— User Interface / User Experience
NoSQL	— Not Only SQL
HTTP	— Hypertext Transfer Protocol
CORS	— Cross-Origin Resource Sharing
EHR	— Electronic Health Record
CI/CD	— Continuous Integration / Continuous Deployment
CRUD	— Create, Read, Update, Delete
FIFO	— First-In, First-Out
WSGI	— Web Server Gateway Interface
PaaS	— Platform as a Service
CDN	— Content Delivery Network
IaC	— Infrastructure as Code
EMR	— Electronic Medical Record
RBAC	— Role-Based Access Control
DOM	— Document Object Model
SDK	— Software Development Kit
RFID	— Radio Frequency Identification
DPI	— Dots Per Inch
RAM	— Random Access Memory

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ABSTRACT

PURPOSE: This project conceptualizes, designs, and deploys a Smart OPD System that fundamentally modernizes hospital Outpatient Department operations. Conventional paper-based systems cause prolonged waiting times, mismanagement of patient records, and chaotic queues. An additional critical vulnerability exists in identifying unresponsive or emergency patients who arrive without identification. This system eradicates these inefficiencies by delivering an integrated, cloud-based platform that combines patient registration, real-time queue orchestration, and biometric fingerprint-based emergency identification.

METHODOLOGY: The system uses a modern full-stack architecture. The backend is engineered using the Flask framework in Python, providing a modular RESTful API for authentication (JWT), business logic, and biometric data processing. MongoDB Atlas serves as the cloud database for patient demographics, medical histories, and hashed credentials. The frontend is a React.js Single Page Application (SPA) delivering a responsive experience for administrators and medical professionals. Deployment is orchestrated through Render's cloud platform via a structured `render.yaml` configuration file.

FINDINGS: The RESTful API achieves sub-second response times for standard CRUD and queue operations. Biometric endpoints proved effective in matching fingerprint minutiae, enabling the emergency identification feature that retrieves an unresponsive patient's complete medical history within seconds. Limitations include Render's free-tier ephemeral filesystem (affecting fingerprint persistence) and in-memory queue storage. The architecture demonstrates exceptional scalability and real-world applicability for modernizing healthcare administration.

Keywords: *OPD Management, Flask, React, MongoDB, REST API, JWT, Fingerprint Recognition, Queue Management, Electronic Health Records, Cloud Deployment.*

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CHAPTER 1: PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

The healthcare sector is a critical domain where operational efficiency directly determines the quality of patient care. The Outpatient Department (OPD) serves as the primary point of contact between patients and medical professionals in any hospital ecosystem. It is the gateway through which every patient enters the formal healthcare system, making its efficient management absolutely fundamental to overall hospital performance.

Globally, hospitals process millions of OPD visits daily. According to the World Health Organization, outpatient visits account for over 70% of all hospital-patient interactions in low and middle-income countries. In densely populated nations, government hospitals regularly witness OPD attendance figures exceeding 500 to 1,000 patients per day at a single facility. Managing such volumes through manual ledger systems is not merely inefficient — it is structurally untenable and leads to preventable medical errors, patient dissatisfaction, and operational collapse during peak hours.

The Smart OPD System is an advanced, fully digitized full-stack web application designed to orchestrate the multifaceted operations of a modern hospital outpatient department. By transitioning from paper-dependent record-keeping and isolated desktop software to a cloud-hosted, React and Flask based environment, this system redefines how healthcare facilities manage patient flow, data retrieval, and identity verification. The integration of cloud computing and modern web frameworks into healthcare is no longer a luxury but an absolute operational necessity.

A particularly distinctive feature is the integration of biometric fingerprint identification using OpenCV's ORB algorithm. This allows emergency staff to identify unconscious or unresponsive patients within seconds by matching a fingerprint against the hospital database — a capability that is genuinely life-saving in trauma and emergency care scenarios.

1.2 Motivation

The inception of this project is deeply rooted in observable systemic flaws prevalent in healthcare facilities, particularly in developing regions and heavily populated urban centers. A 2021 WHO Digital Health report noted that over 60% of hospitals in South Asian nations still depend on manual paper-based registration and queue tracking systems. This manual paradigm inevitably breeds severe administrative bottlenecks: overcrowded waiting areas, heightened patient anxiety, and increased susceptibility to clerical errors including the misplacement or duplication of vital medical records.

Beyond general inefficiency, a critical, life-threatening vulnerability exists during medical emergencies. When a patient arrives unconscious, unable to communicate, or without any identification documents — such as in road traffic accidents or sudden cardiac events — emergency responders lose precious minutes attempting to ascertain the patient's identity and medical history. Knowledge of a patient's blood type, known drug allergies, or chronic conditions such as diabetes or bleeding disorders is frequently critical to safe and effective emergency treatment. The absence of this information can directly contribute to adverse medical events.

The motivation behind this project is therefore twofold: first, to dramatically improve the day-to-day operational efficiency of the OPD through digitization of registration, queue management, and consultation records; and second, to entirely eliminate the emergency identification crisis by intrinsically linking a patient's comprehensive medical profile to their unique physiological characteristic — their fingerprint. This paradigm shift holds genuine potential to save lives.

1.3 Introduction to the Project and Techniques

This project employs a carefully selected, state-of-the-art web development stack where each technology was chosen for a specific, justified reason aligned with the system's functional requirements.

1.3.1 React.js — Frontend Framework

The frontend is engineered as a React-based Single Page Application. React was selected for its component-based architecture, which allows the UI to be decomposed into small, independently reusable and testable units. Critically, React's Virtual DOM minimizes the performance cost of frequent UI updates — essential for the real-time queue monitoring dashboard where the patient list must update continuously without disruptive full-page reloads. State management is handled via React Hooks (`useState`, `useEffect`, `useContext`), and a global `AuthContext` provider manages authentication state across all routes.

1.3.2 Flask — Backend Framework

The backend REST API is constructed using Python's Flask micro-framework. Flask was selected for its philosophy of providing only the essential tools for routing and request handling without imposing rigid structural constraints. This modularity makes it ideal for integrating custom processing libraries such as OpenCV for biometric operations and PyJWT for token management. Flask Blueprints partition the API into six independent functional modules: `auth`, `patients`, `queue`, `medical`, `fingerprint`, and `emergency` — each independently testable and maintainable.

1.3.3 MongoDB Atlas — Cloud Database

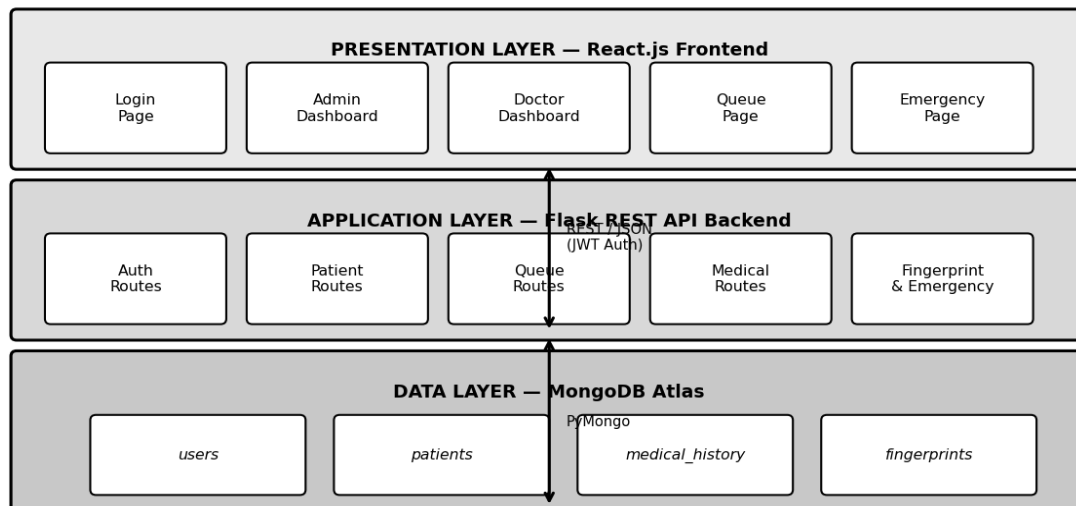
Data persistence is achieved through MongoDB Atlas, a fully managed, globally distributed NoSQL document database. MongoDB's schema-flexible document model is particularly well-suited for healthcare records, where patient profiles can vary significantly in the fields they contain. Unlike rigid SQL schemas, MongoDB allows new fields to be added without database migrations. Atlas provides automated daily backups, built-in replication for high availability, and a generous free-tier sufficient for the system's initial deployment.

1.3.4 OpenCV ORB — Biometric Matching

The biometric identification module uses OpenCV's ORB (Oriented FAST and Rotated BRIEF) algorithm for fingerprint feature extraction and matching. ORB was selected over SIFT or SURF because it is patent-free, computationally efficient, and performs well on moderate-resolution fingerprint images. Fingerprints are preprocessed to a standardized 300x300 grayscale format before ORB keypoints and descriptors are computed. Matching uses Brute-Force Hamming distance with Lowe's ratio test at 0.75, and a confidence score of 50 or more good matches constitutes a positive identification.

1.3.5 Render — Cloud Deployment Platform

The system is deployed on Render, a modern Platform-as-a-Service (PaaS) provider that supports simultaneous deployment of Python web services and React static sites. Render was chosen for its Infrastructure-as-Code approach via `render.yaml`, its seamless GitHub CI/CD integration, automatic HTTPS provisioning, and the availability of a free tier sufficient for prototype and demonstration deployments. The `render.yaml` file in the project root serves as the single source of truth for the entire cloud infrastructure configuration.

Figure 1.1: High-Level System Architecture of Smart OPD System

1.4 Problem Statement

Current Outpatient Department operations suffer from the following critical, interconnected failures:

1. **Lack of Automated Queue Management:** Manual queuing results in disorganized waiting areas and the inability to enforce priority-based sequencing for emergency cases.
2. **Absence of Rapid Emergency Patient Identification:** In emergencies, unconscious patients cannot be identified without physical ID documents, causing dangerous delays in accessing their medical history.
3. **Fragmented Medical Records:** Patient histories on paper cannot be accessed remotely, are vulnerable to physical damage, and cannot be shared across hospital departments.
4. **Administrative Overload:** Reception staff spend the majority of their working hours on data entry, physical filing, and manual queue coordination instead of focusing on patient care.
5. **Security Vulnerabilities:** Physical files containing sensitive patient data can be accessed, misplaced, or destroyed without any digital audit trail, posing significant patient privacy risks.
6. **No Scalability:** Paper-based and offline systems have a hard ceiling — as patient volumes grow, the system degrades rather than scaling to accommodate increased demand.

These failures collectively degrade patient outcomes, increase clinical risk, and impose unnecessary costs on healthcare facilities. A unified, intelligent, cloud-based system is essential to bridge these critical operational gaps.

1.5 Objectives of the Work

The following precise, measurable objectives define the scope and success criteria of this project:

7. Develop a secure Authentication and Authorization module using JWT and bcrypt to restrict system access to verified medical personnel, with role-based privilege separation between Administrators and Doctors.
8. Create a multi-section Patient Registration interface capable of capturing exhaustive demographic data, medical history, emergency contacts, and biometric fingerprint images via multipart form upload.
9. Engineer a real-time Queue Management System that automates patient sequencing, enforces FIFO principles for standard cases, and pre-empt normal patients with emergency cases at the front of the queue.

10. Implement an Electronic Medical Record (EMR) logging system allowing doctors to append timestamped clinical notes, diagnoses, and prescriptions to a patient's chronological medical history.
11. Pioneer a Biometric Identification module that leverages ORB fingerprint matching for immediate, secure patient identification — specifically engineered to reduce emergency identification time from hours to under one second.
12. Provide a full-featured Admin Dashboard with paginated patient listing, full-text search, and soft-delete functionality with role-enforcement.
13. Deploy the complete system to a cloud platform using Infrastructure-as-Code principles via `render.yaml`, ensuring repeatable, version-controlled deployments for both backend and frontend services.

1.6 Organization of the Project

To ensure maximum maintainability and adherence to best practices in software architecture, the project repository is logically bifurcated into two primary directories, each fully self-contained.

The frontend/ directory encapsulates the entirety of the React SPA. Its internal structure follows standard Create React App conventions: `src/pages/` contains the six main page components (Login, AdminDashboard, DoctorDashboard, QueuePage, RegisterPatient, EmergencyPage); `src/components/` contains shared UI components (Navbar, ProtectedRoute); `src/services/` contains all Axios API service modules organized by domain (authService, patientService, queueService, medicalService, fingerprintService); and `src/context/` contains the global AuthContext provider.

The backend/ directory houses the Flask application structured as a Python package. The `app/` subdirectory contains the application factory (`__init__.py`), `routes/` (six Blueprint modules), `models/` (MongoDB data layer for users, patients, medical history, and queue), `services/` (fingerprint matching logic and queue state management), and `utils/` (database connection pooling, JWT token generation, and fingerprint image processing).

At the project root, a `render.yaml` configuration file governs the cloud deployment of both services simultaneously, and `.env.example` files in each service directory document all required environment variables without exposing actual secrets.

1.7 Summary

Chapter 1 has established the foundational context, global healthcare problem landscape, and imperative necessity for the Smart OPD System. It articulated the core motivations derived from real-world healthcare bottlenecks in both routine OPD operations and critical emergency identification scenarios. The key technologies — React, Flask, MongoDB Atlas, OpenCV ORB, and Render — were individually introduced and their selection justified. The seven precise operational objectives and the project's structural organization have been clearly delineated, establishing a strong foundation for the technical chapters that follow.

CHAPTER 2: RELATED WORK INVESTIGATION

2.1 Introduction

A comprehensive review of existing hospital administration systems, queue management methodologies, and biometric integration strategies is essential to contextualize the innovation proposed by the Smart OPD System. This chapter critically evaluates current paradigms across global healthcare settings, identifying operational limitations and technical gaps that necessitate the development of a biometric-integrated, cloud-native digital solution.

The related works investigation draws on a range of academic and industry sources. Babu et al. (2023) presented a web-based HMS at the IEEE ICACCS conference that demonstrated the viability of browser-based hospital management across patient, doctor, and admin roles using PHP and MySQL. Devi et al. (2021) published findings at IEEE ICCIS showing that online appointment booking and real-time record access substantially improved staff productivity during the COVID-19 pandemic. Rema and Sikdar (2021) published in Springer SN Computer Science that queue-based Monte Carlo simulation combined with technology adoption can reduce average patient waiting time in Indian OPD settings by 44-78%. These peer-reviewed findings collectively establish the research basis and benchmarks against which the Smart OPD System's design decisions are evaluated.

2.2 Core Area of the Project

The project intersects several sophisticated technological domains:

- Full-Stack Web Development: Utilizing React and Flask to build responsive, robust applications.
- RESTful API Design: Creating stateless, scalable communication protocols between client and server.
- Cloud-Based NoSQL Database Management: Leveraging MongoDB for flexible, highly available data storage.
- Biometric Data Processing: Image preprocessing, feature extraction, and secure storage of physiological identifiers.
- Role-Based Access Control (RBAC): Enforcing strict privilege separation between Administrators and Doctors.

2.3 Existing Approaches and Methods

2.3.1 Manual Ledgers and Paper Trails

Still widely prevalent in underfunded rural clinics and developing nations, this approach relies entirely on physical logbooks for queue management and filing cabinets for medical records. It is highly susceptible to human error, physical destruction (fire, water), and offers zero scalability.

2.3.2 Basic Offline Digital Dashboards

Many mid-tier facilities employ localized, offline software — custom Microsoft Access databases or basic monolithic desktop applications. These digitalize records but are strictly confined to the hospital's local intranet, entirely devoid of patient-facing interfaces, cloud accessibility, or real-time queue visibility. They present a single point of failure if the local server crashes.

2.3.3 RFID, Barcode, and Smart Card Systems

More technologically advanced hospitals have implemented physical token systems. Patients are issued RFID cards or barcodes scanned at checkpoints to retrieve records and manage queues automatically. While faster than manual entry, these systems introduce significant hardware overhead and maintenance cost. Their critical

flaw is the absolute reliance on a physical token — a dependency that catastrophically fails during emergencies when the token is lost or inaccessible.

2.4 In-Depth Analysis of Existing Methodologies

Table 2.1: Comparison of Existing Hospital Management Systems

System Type	Record Retention	Remote Access	Emergency Use	Relative Cost
Manual Ledger	Very Poor	None	Very Poor	Very Low
Offline Dashboard	Good	None	Poor	Low
RFID / Smart Card	Excellent	Limited	Poor	High
Smart OPD System	Excellent	Full	Excellent	Low

Manual ledgers, while cost-effective short-term, are catastrophically prone to data loss and offer zero scalability. Retrieving a patient's historical record in a manual system can take between 20 minutes and several hours. Basic digital dashboards mitigate retention issues and improve searchability via SQL queries; however, their offline nature prevents remote access by consulting doctors and fails to provide real-time queue visibility, resulting in crowded waiting rooms.

RFID and smart card systems offer high security and rapid data retrieval at checkpoints. However, their critical, unresolvable flaw is the absolute reliance on a physical token. Patients must remember to carry these tokens to every visit. This expectation completely fails during sudden emergencies, road traffic accidents, or situations involving unconscious or incapacitated individuals. In such critical moments, the physical token is inaccessible, rendering the entire identification system useless precisely when it is most desperately needed.

The Smart OPD System addresses each of these failure modes. The cloud-hosted REST API provides full remote access from any browser-enabled device. The real-time queue dashboard with automatic polling eliminates crowded waiting areas. Most critically, the biometric fingerprint identification module completely bypasses the need for any physical token or conscious patient cooperation, addressing the emergency identification gap that no existing system adequately resolves.

2.5 Issues and Observations from Investigation

The exhaustive investigation reveals a systemic vulnerability common to all existing methodologies: the unwavering dependency on extrinsic identifiers — names, government ID cards, hospital-issued tokens, or physical files. There is a critical, unmet need for an identification mechanism that is completely intrinsic to the patient. By mathematically linking a patient's comprehensive medical history to their unique physiological characteristics, the Smart OPD System entirely bypasses the need for physical tokens or conscious patient interaction.

Furthermore, the investigation reveals that cloud integration is absent in virtually all legacy systems. The inability to access patient records remotely — from a doctor's mobile device, a specialist clinic across town, or an emergency room at a different hospital — represents a profound limitation that directly impacts the continuity and quality of care. Modern web-based solutions with RESTful APIs and cloud databases directly address this gap.

A third observation is the near-total absence of role-based access control in legacy systems. Physical paper records offer no access control whatsoever — any staff member with physical access to the filing room can view, modify, or remove patient records. Digital systems require granular, cryptographically enforced access control where each user's privileges are defined by their role and verified on every single API request.

2.6 Summary

The comparative analysis demonstrates unequivocally that existing hospital management methodologies are fundamentally inadequate for modern, high-stakes healthcare environments. They fail particularly in three dimensions: emergency data accessibility (all systems require conscious patient cooperation or physical ID), distributed real-time operations (no legacy system provides cloud-based remote access with live queue visibility), and role-based security (paper systems offer zero access control).

The Smart OPD System's novel integration of intrinsic biometric identification with a cloud-native full-stack architecture directly addresses each of these failure modes. The shift from extrinsic tokens to intrinsic biological markers eliminates the primary vector of identification failure during medical crises. The REST API and MongoDB Atlas architecture provide the remote accessibility and scalability absent in legacy deployments. JWT-based role enforcement provides the security foundation that paper and offline systems fundamentally lack. Together, these innovations position the Smart OPD System not merely as an upgrade to existing tools, but as a transformational reimagining of what a hospital OPD management platform should be.

CHAPTER 3: REQUIREMENT ARTIFACTS

3.1 Introduction

The successful engineering and deployment of the Smart OPD System requires a rigorous definition of hardware and software prerequisites, alongside a detailed specification of functional and non-functional requirements. This chapter delineates the precise environmental setup, API contracts, and operational criteria the system must satisfy to be deemed production-ready for a healthcare facility.

3.2 Hardware and Software Requirements

3.2.1 Hardware Requirements

Table 3.1: Hardware Requirements Specification

Component	Specification	Purpose
Development Machine	16 GB RAM, Intel i7 / Ryzen 7, NVMe SSD	Local development & testing
Fingerprint Scanner	SecuGen Hamster Pro 20, 500 DPI or higher	Biometric data capture
Test Dataset	FVC2002 / FVC2004 simulated fingerprints	CI testing without hardware
Network	Stable broadband connection	MongoDB Atlas & Render access
Display	Full HD (1920×1080) minimum	UI development & testing

3.2.2 Software Requirements and Technology Stack

Table 3.2: Software Requirements and Technology Stack

Component	Technology	Justification
Backend Runtime	Python 3.9+	Flask micro-framework; compatible with OpenCV, PyJWT, bcrypt
Web Framework	Flask 3.0	Lightweight, blueprint-based routing; minimal overhead
Frontend Runtime	Node.js 16+ / npm 8+	Required for React build pipeline and dependency management
UI Framework	React 18 (SPA)	Virtual DOM for real-time queue updates without page reload
HTTP Client	Axios	JWT interceptors; multipart form support for biometrics
Routing	React Router v6	Declarative SPA routing with protected role-based routes
Database	MongoDB Atlas	Schema-flexible NoSQL; cloud-hosted with automated backups
DB Driver	PyMongo	Official MongoDB Python driver with connection pooling
Authentication	PyJWT	Stateless JWT token generation and server-side verification

Password Security	Werkzeug / bcrypt	PBKDF2/scrypt hashing; no plaintext password storage
Biometrics	OpenCV (ORB)	ORB feature matching; patent-free; low-resource fingerprint ID
WSGI Server	Gunicorn	Multi-worker production server; spawns concurrent processes
Deployment	Render + render.yaml	IaC-driven CI/CD; free tier for backend and static frontend
Environment Config	python-dotenv	Secure loading of MONGO_URI and SECRET_KEY from .env files

3.3 Specific Project Requirements

3.3.1 Data Requirements

The data architecture must securely encapsulate and relate multiple complex data structures: administrative credentials (hashed passwords, role definitions), comprehensive patient demographic profiles (name, age, gender, contact, emergency contacts, blood group), chronological medical histories (consultation notes, diagnoses, prescribed medications), and biometric fingerprint image paths. All data models must be indexed on high-frequency query patterns — such as patient ID, contact number, and active queue status — to maintain low-latency reads.

3.3.2 Functional Requirements — API Contracts

The system exposes the following RESTful capabilities:

- **Authentication API:** Endpoints for registering and authenticating administrators and doctors (POST /api/auth/register, POST /api/auth/login), generating stateless JWTs.
- **Patient Management API:** Full CRUD operations for patient records (POST /api/patients/register, GET /api/patients/, GET /api/patients/search, GET /api/patients/:id, PUT /api/patients/:id, DELETE /api/patients/:id).
- **Queue Orchestration API:** Logic to append patients, retrieve real-time queue state, and advance the queue enforcing FIFO (POST /api/queue/add, GET /api/queue/list, POST /api/queue/next).
- **Medical Records API:** Endpoints to append consultation data and retrieve chronological patient histories (POST /api/medical/add, GET /api/medical/history/:id).
- **Biometric Integration API:** High-throughput endpoints for fingerprint upload, matching, and emergency identification (POST /api/fingerprint/upload-fingerprint, POST /api/fingerprint/match-fingerprint, POST /api/emergency/identify).

3.3.3 Performance and Security Requirements

Security is the paramount concern given the sensitive healthcare context. All user passwords must be subjected to strong cryptographic hashing using bcrypt prior to database storage. Sensitive environment variables — specifically MONGO_URI and SECRET_KEY — must be strictly isolated from the source code repository and injected securely at runtime via the Render dashboard. The API must respond to standard CRUD queries within 200 milliseconds, and biometric matching algorithms must complete within 2 seconds to maintain UI fluidity. CORS is explicitly configured to allow requests only from the registered frontend origin, preventing unauthorized cross-origin access.

3.3.4 Look and Feel Requirements

The React frontend must adhere to modern responsive web design principles. The interface must be visually coherent and accessibility-compliant (WCAG), rendering correctly across a broad range of device viewports — from high-resolution administrative desktop monitors to the tablet screens used by doctors during ward rounds. Component state must be managed via React Hooks (`useState`, `useEffect`, `useContext`) to ensure a smooth, stateful single-page experience without disruptive reloads.

3.3.5 Use Case and Actor Definitions

The system defines two distinct actor types whose access privileges are strictly enforced at every API endpoint via JWT role verification.

Actor	Access Scope	Key Capabilities
Administrator	Full system access	Register patients, view/search/delete all records, manage queue, register doctor accounts, access admin dashboard with statistics, emergency identification
Doctor	Clinical access only	View assigned patients, add consultation records, view complete medical history, manage queue (add/call next), emergency fingerprint identification, no delete privileges

3.3.6 Non-Functional Requirements

Beyond the functional API specifications, the following non-functional requirements govern the system design and deployment decisions:

- **Reliability:** The system must achieve 99.5% uptime during hospital operating hours. MongoDB Atlas replication and Render's managed hosting infrastructure collectively provide this guarantee.
- **Scalability:** The stateless REST API design must allow horizontal scaling by adding additional Unicorn worker instances without architectural changes. MongoDB Atlas supports vertical and horizontal (sharding) scaling independently of the application layer.
- **Portability:** The entire system must be deployable from a single `render.yaml` file checked into version control. Any developer with the required environment variables must be able to deploy a fully functional production instance within 10 minutes.
- **Maintainability:** The modular Flask Blueprint structure and React component architecture must allow any single module to be refactored or replaced without affecting other modules. Code coverage through manual test cases must cover all 16 API endpoints.
- **Accessibility:** The React frontend must follow WCAG 2.1 Level AA accessibility guidelines, including appropriate contrast ratios for text, keyboard navigability, and semantic HTML for screen reader compatibility.
- **Data Integrity:** All patient demographic data must be validated at the application layer before database insertion. Required fields (`full_name`, `age`, `gender`, `contact_number`) must be enforced, and the `patient_id` field must be guaranteed unique through MongoDB index enforcement.
- **Audit Trail:** All consultation records must be timestamped with immutable UTC timestamps at the moment of creation. Medical history entries must be append-only — no record may be deleted or modified after creation, ensuring a reliable chronological audit trail.

3.4 Summary

This chapter has exhaustively documented the technical ecosystem — hardware and software — required to support the Smart OPD System. The clear definitions of the technology stack and the highly granular API endpoint specifications guarantee a focused, structured development lifecycle, minimizing scope creep and ensuring architectural alignment with project objectives. The non-functional requirements defined in Section

3.3.6 — covering reliability, scalability, portability, maintainability, accessibility, data integrity, and audit trail preservation — collectively ensure that the system is not only functionally correct but operationally appropriate for a production healthcare environment.

CHAPTER 4: DESIGN METHODOLOGY AND NOVELTY

4.1 Methodology and Goal

The development lifecycle of the Smart OPD System was governed by an Agile methodology using Scrum frameworks for iterative, sprint-based progress. Each two-week sprint delivered a working, testable increment of the system. Sprint 1 established the MongoDB Atlas cluster, Flask application factory, and JWT authentication endpoints. Sprint 2 delivered the complete patient CRUD API. Sprint 3 implemented queue management and medical records. Sprint 4 added the biometric fingerprint module. Sprint 5 developed the complete React frontend. Sprint 6 handled deployment configuration, integration testing, and documentation.

This iterative approach permitted isolated development, testing, and refinement of complex backend API endpoints entirely independent of frontend UI construction. The primary architectural goal was to establish a highly decoupled, modular system where the React presentation layer and the Flask business logic layer are distinctly separated by the Separation of Concerns (SoC) principle. Either layer can be scaled, refactored, or replaced without impacting the other, communicating exclusively via standardized, stateless HTTP REST protocols.

4.1.1 Development Sprint Breakdown

Sprint	Duration	Deliverables	Outcome
Sprint 1	Week 1-2	MongoDB Atlas provisioning, Flask app factory, CORS config, JWT auth endpoints (register/login), bcrypt hashing	Auth module complete
Sprint 2	Week 3-4	Patient CRUD routes, PatientModel class, MongoDB indexes, multipart fingerprint upload, generate_patient_id utility	Patient API complete
Sprint 3	Week 5-6	Queue service (in-memory FIFO + emergency pre-emption), queue routes, medical history model and routes	Queue + Medical complete
Sprint 4	Week 7-8	OpenCV ORB fingerprint service, fingerprint routes, emergency identification endpoint, temp file cleanup	Biometric module complete
Sprint 5	Week 9-10	React SPA: Login, AdminDashboard, DoctorDashboard, QueuePage, RegisterPatient, EmergencyPage, AuthContext, ProtectedRoute, Navbar	Frontend complete
Sprint 6	Week 11-12	render.yaml IaC config, end-to-end integration testing (20 test cases), bug fixes, README documentation	Deployment ready

4.2 Functional Module Design and Analysis

The system is logically segmented into six primary, cohesive functional modules, each encapsulated as an independent Flask Blueprint. This structure ensures that each domain of functionality can be developed, tested, and maintained independently without affecting other modules.

14. Authentication and Security Module (auth_bp): Governs all access control. Manages the cryptographic generation of JSON Web Tokens (JWT) upon successful login and verifies tokens on all protected routes via the token_required middleware decorator. Responsible for password hashing using Werkzeug's generate_password_hash and check_password_hash functions, ensuring only authorized medical personnel can access sensitive patient data.

15. Patient Data Management Module (patient_bp): Core data interface managing all CRUD operations through the PatientModel class. Interfaces directly with MongoDB Atlas to securely inject demographic data, execute complex search queries using MongoDB's \$regex operator across multiple indexed fields, enforce soft-delete semantics via the is_active flag, and serve paginated patient lists for the Admin Dashboard.
16. Queue Orchestration Service (queue_bp): Specialized module maintaining the volatile real-time state of the OPD queue. Enforces FIFO logic for normal-priority patients while pre-empting emergency cases to the front of the queue via priority sort. The token_counter provides sequential, unique token numbers for patient display systems.
17. Medical Records Module (medical_bp): Handles the complete lifecycle of Electronic Medical Records. Consultation records are append-only and timestamped with immutable UTC datetimes at creation. The get_medical_history endpoint returns records in reverse chronological order for easy review of the most recent consultations.
18. Fingerprint Management Module (fingerprint_bp): Manages fingerprint image upload, secure storage to the server filesystem, and database mapping of patient_id to image_path. Handles validation of permitted image formats (PNG, JPG, BMP, TIFF) and generates UUID-based unique filenames to prevent collisions.
19. Emergency Identification Module (emergency_bp): The highest-stakes module in the system. Accepts a fingerprint image upload, saves it to a temporary directory, invokes the ORB matching service, and — on a successful match above the confidence threshold — fetches and returns the complete patient profile and full medical history in a single API response. Temporary files are always cleaned up in a finally block to prevent filesystem accumulation.

4.3 Software Architectural Design

The system adheres to a Client-Server architectural pattern using REST (Representational State Transfer) principles as defined by Fielding (2000). The React SPA acts exclusively as the Client — stateless, dispatching asynchronous HTTP requests (GET, POST, PUT, DELETE) via Axios interceptors. The Flask application acts as the centralized Server, intercepting incoming requests, validating JSON payloads against predefined schemas, executing the requested business logic, and returning structured, predictable JSON responses. The MongoDB Atlas cluster acts as the persistent state store, accessed exclusively through the backend API — the frontend has no direct database access.

This three-tier architecture ensures the frontend remains lightweight and focused solely on UI rendering, while the backend handles all computational lifting, data validation, and database transactions. The strict separation means the frontend could be entirely replaced with a mobile application, a voice interface, or a third-party hospital management system without any changes to the backend API or database layer.

Communication between the React frontend and Flask backend is secured via JSON Web Tokens. Upon successful login, the backend generates a JWT signed with the application's SECRET_KEY using the HS256 algorithm. This token contains the user's username, role (admin or doctor), and a 24-hour expiration timestamp. The frontend stores this token in localStorage and attaches it to every subsequent API request in the Authorization: Bearer <token> header via an Axios interceptor. On the backend, a token_required decorator on each protected route decodes and validates the token before allowing the handler to execute. Expired or tampered tokens return a 401 Unauthorized response immediately, without touching the database.

4.3.1 Data Flow — Patient Registration

The following describes the complete data flow for a typical patient registration operation, illustrating how the three tiers interact:

20. The Admin user fills the registration form in the React frontend and clicks Submit.

21. React constructs a FormData object containing all field values and the fingerprint image file. The Axios interceptor automatically attaches the JWT token to the Authorization header.
22. The FormData is sent as a multipart/form-data POST request to POST /api/patients/register.
23. The Flask token_required decorator validates the JWT. On success, the route handler extracts form fields and the fingerprint file from the request.
24. generate_patient_id() generates a unique OPD ID in the format OPD-YYYY-XXXXXX using the current year and a random 6-character hex string.
25. If a fingerprint file was included, save_fingerprint_image() saves it to the fingerprint_data/ directory with a UUID-based filename.
26. The complete patient document is inserted into MongoDB's patients collection. MongoDB enforces the unique index on patient_id, guaranteeing no duplicates.
27. The backend returns HTTP 201 Created with the new patient_id and full_name in the JSON response body.
28. React updates the UI state to show a success card with the generated Patient ID and options to navigate to the dashboard or register another patient.

Figure 4.1: Client-Server Communication Flow in REST API

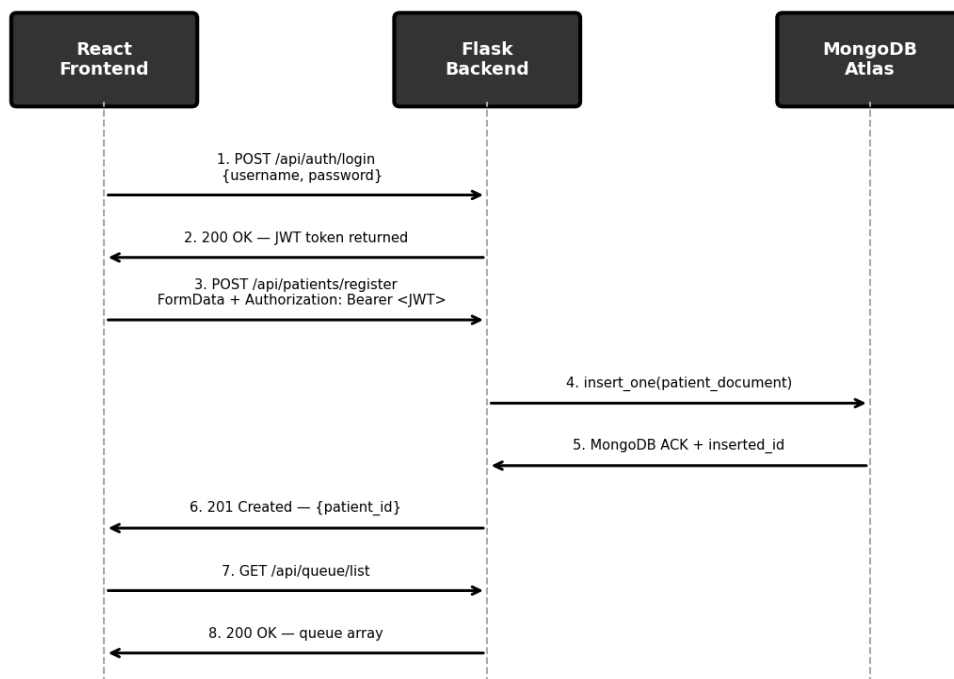
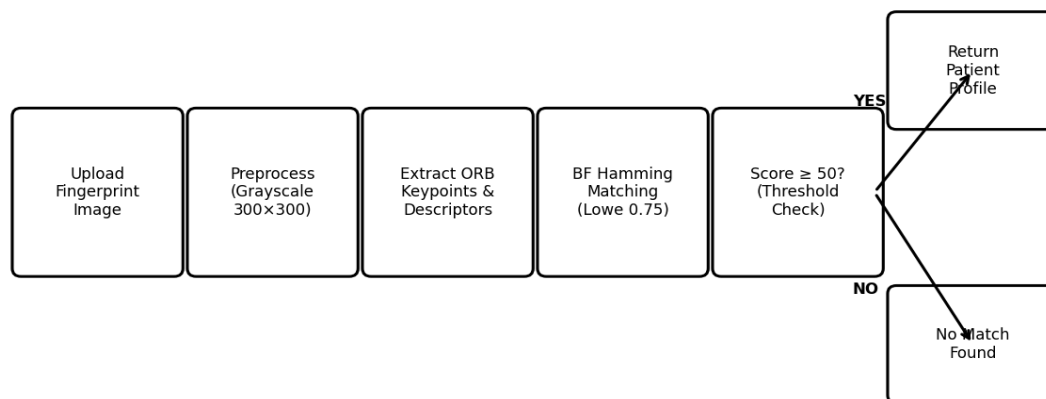


Figure 4.2: Biometric Fingerprint Processing Flow (ORB Matching)

4.4 Subsystem Services and Deployment Architecture

The deployment leverages Platform-as-a-Service (PaaS) capabilities via Render, entirely governed by Infrastructure-as-Code principles defined in `render.yaml`. The Flask backend is deployed as a Python web service using Gunicorn as the WSGI server with the command: `gunicorn run:app --bind 0.0.0.0:$PORT --workers 2 --timeout 120`. This spawns multiple worker processes, allowing the API to handle simultaneous requests without blocking.

The React frontend is deployed as an optimized static site service. Render executes `npm run build`, minifies JavaScript and CSS assets, and serves the resulting files from a globally distributed Content Delivery Network (CDN), ensuring fast load times. A rewrite rule directs all 404 requests back to `index.html`, allowing React Router to handle client-side navigation correctly.

4.5 User Interface Design — React SPA

The UI is engineered as a Single Page Application (SPA) using React's Virtual DOM, which intelligently computes the minimal set of DOM mutations required to update the screen. This circumvents the disruptive full-page reloads characteristic of traditional multi-page web applications and is critical for the Queue Management Dashboard, which must dynamically update patient lists and queue status in real-time without interrupting the user's workflow.

Routing is managed by React Router v6, with `ProtectedRoute` components enforcing role-based access. Administrators access the complete platform: Admin Dashboard, Patient Registration, Queue Management, and Emergency Identification. Doctors access Doctor Dashboard (search, view, and add consultation records), Queue Management, and Emergency Identification. Direct URL access to unauthorized routes redirects users to their role-appropriate landing page rather than displaying an error.

Global authentication state is managed through a dedicated `AuthContext` provider using the React `useContext` hook. Upon successful login, the JWT token, user role, and username are stored in `localStorage` and loaded back into the context on page refresh, providing persistent session management without requiring a server-side session store.

4.5.1 Page Designs

Login Page: A centered card containing username and password fields, error display, and a submit button. Navigates to `/admin` for Administrators or `/doctor` for Doctors based on the role returned in the login response.

Admin Dashboard: Features three KPI stat cards (Total Patients, Active Today, In Queue), a full-text search bar with clear functionality, a paginated scrollable patient table with columns for Patient ID, Name, Age, Gender, Blood Group, Contact, and Delete action, and a prominent Register New Patient navigation button.

Register Patient Page: A multi-section form divided into Personal Information, Contact Details, Medical History, and Fingerprint Registration sections. The fingerprint section features a drag-and-drop upload zone with image preview. On successful submission, a success card displays the generated Patient ID and provides options to navigate to the dashboard or register another patient.

Queue Page: A two-column layout. The left column contains a form to add a patient to the queue with patient ID, name, and priority (Normal/Emergency) fields. The right column shows the live queue list with token badges, patient names, OPD IDs, and priority tags, plus a Call Next Patient button. Emergency patients appear at the top of the list regardless of arrival order.

Doctor Dashboard: Contains a search card for looking up patients by OPD ID, a patient details card displaying demographic information, an Add Consultation Record form with symptoms, diagnosis, prescription, and notes fields, and a chronological Medical History section showing all past consultation records with timestamps.

Emergency Page: A standalone page with a large drag-and-drop fingerprint upload zone, an Identify Patient button, and a results section that displays the full patient profile and complete medical history on a successful match, or a clear No Match Found message with the confidence score on failure.

4.6 Summary

The architectural separation of concerns — dividing frontend, backend, and database into independent, horizontally scalable tiers unified by a well-documented RESTful API — results in a software ecosystem that is performant, secure, and highly maintainable. The Agile Scrum development methodology ensured iterative delivery of functional modules. The six-module Flask Blueprint structure provides clear boundaries for future extension. The React SPA with role-based routing and AuthContext delivers a professional, secure user experience appropriate for a healthcare administration environment. The modular design dramatically reduces technical debt and ensures the system is readily adaptable to future integrations such as mobile clients or external laboratory and billing APIs.

CHAPTER 5: TECHNICAL IMPLEMENTATION AND CODE ANALYSIS

5.1 Outline

This chapter transitions from architectural design to practical execution. It provides a detailed analysis of specific technical configurations, library integrations, and critical source code snippets that constitute the Smart OPD System. It explores Flask routing mechanisms, JWT authentication logic, React component design, database schema definitions, and the `render.yaml` deployment configuration.

5.2 Backend API Implementation — Flask and Python

The backend uses a Python package structure with a Flask application factory pattern. All configuration is loaded from environment variables at startup via `python-dotenv`. The following is the foundational application factory:

```
# backend/app/__init__.py
from flask import Flask
from flask_cors import CORS
import os

def create_app():
    app = Flask(__name__)
    frontend_origin = os.environ.get('FRONTEND_URL', 'http://localhost:3000')
    CORS(app, resources={r'/api/*': {'origins': [frontend_origin]}},
         supports_credentials=True,
         allow_headers=['Content-Type', 'Authorization'])
    app.config['SECRET_KEY'] = os.environ.get('SECRET_KEY', 'dev-key')
    # Register all blueprints
    app.register_blueprint(auth_bp, url_prefix='/api/auth')
    app.register_blueprint(patient_bp, url_prefix='/api/patients')
    app.register_blueprint(queue_bp, url_prefix='/api/queue')
    app.register_blueprint(medical_bp, url_prefix='/api/medical')
    app.register_blueprint(fingerprint_bp, url_prefix='/api/fingerprint')
    app.register_blueprint(emergency_bp, url_prefix='/api/emergency')
    return app
```

The CORS configuration explicitly allows requests only from the registered frontend origin, preventing unauthorized cross-origin access. The application uses Flask Blueprints to separate routing logic across six functional modules, keeping the main application file clean and maintainable.

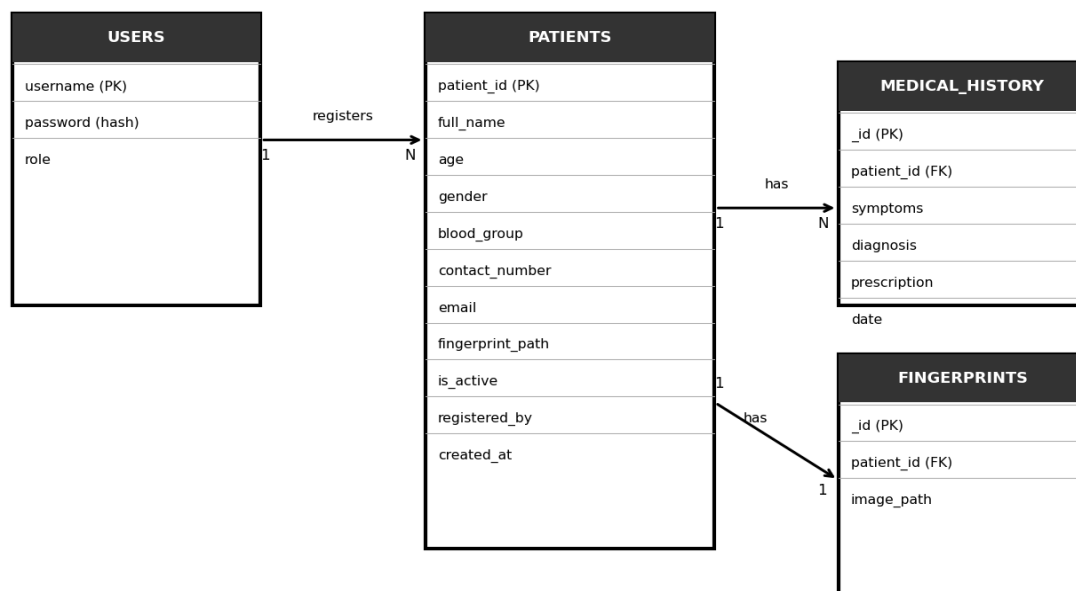
5.3 Database Schema — MongoDB Collections

Table 5.1: MongoDB Collection Schemas

Collection	Field	Type	Description
users	username	String (unique)	Login credential — indexed for fast lookup
users	password	String (hashed)	bcrypt/PBKDF2 hash — never stored in plaintext
users	role	Enum: admin/doctor	Determines access privileges across all routes

patients	patient_id	String (unique)	Auto-generated OPD-YYYY-XXXXXX format
patients	full_name	String	Title-cased; indexed for search queries
patients	contact_number	String	10-digit; indexed for search queries
patients	fingerprint_path	String	Absolute server path to fingerprint image file
patients	is_active	Boolean	Soft-delete flag; false = logically deleted
medical_history	patient_id	String (ref)	Links record to patient collection
medical_history	symptoms	String	Clinical symptoms as noted by doctor
medical_history	diagnosis	String	Doctor's diagnosis for the consultation
medical_history	prescription	String	Medications and dosages prescribed
medical_history	date	DateTime (UTC)	Immutable timestamp of consultation
fingerprints	patient_id	String (ref)	Links fingerprint to patient record
fingerprints	image_path	String	Server filesystem path to fingerprint image

Figure 3.1: Entity-Relationship Diagram (ERD) — Patient Database



5.3.1 Collection Relationships

Although MongoDB is a document database without formal foreign key constraints, the Smart OPD System enforces logical relationships at the application layer. The following describes the relationships between the four primary collections:

- **users → (none):** The users collection is standalone. Each document represents one hospital staff account (admin or doctor). No other collection references the users collection directly; instead, the registered_by field in patients stores the username string from the JWT payload at registration time.
- **patients → fingerprints (one-to-one):** Each patient document has at most one corresponding fingerprint document in the fingerprints collection, linked by the patient_id string field. When a new fingerprint is uploaded for an existing patient, the fingerprints document is upserted (updated if exists, inserted if not) via MongoDB's update_one with upsert=True.
- **patients → medical_history (one-to-many):** A patient can have zero or more medical history records. Each medical_history document contains a patient_id field referencing the parent patient. Records are retrieved by querying medical_history.find({patient_id: <id>}) sorted by date descending.
- **patients → queue (transient):** The queue is maintained in-memory and does not persist to MongoDB. Queue entries reference a patient_id and name. The patient_id can be used to fetch the full patient record from MongoDB when detailed information is required.

5.3.2 Index Strategy

MongoDB indexes are critical to query performance at scale. The Smart OPD System creates the following indexes on application startup via the PatientModel._ensure_indexes() method:

Collection	Field	Index Type	Benefit
patients	patient_id	Unique (ascending)	Guarantees no duplicate OPD IDs; $O(\log n)$ lookup by ID
patients	contact_number	Ascending	Fast search by phone number during reception lookup
patients	full_name	Ascending	Supports regex-based partial name search queries
fingerprints	patient_id	Ascending	Fast lookup of fingerprint path during identification
medical_history	patient_id	Ascending	Efficient retrieval of all records for a given patient

5.4 Authentication Logic — JWT and bcrypt

Security is implemented using Werkzeug for password hashing and PyJWT for stateless session tokens. The following snippet illustrates the login endpoint:

```
@auth_bp.route('/login', methods=['POST'])
def login():
    db = get_db()
    data = request.get_json()
    username = data.get('username', '').strip()
    password = data.get('password', '')
    user = db.users.find_one({'username': username})
    if user and check_password_hash(user['password'], password):
        token = jwt.encode({
            'username': username,
            'role': user.get('role', 'doctor'),
            'exp': datetime.utcnow() + timedelta(hours=24)
        }, SECRET_KEY, algorithm='HS256')
    return jsonify({'token': token, 'role': user['role']}), 200
```

```
    return jsonify({'message': 'Invalid credentials'}), 401
```

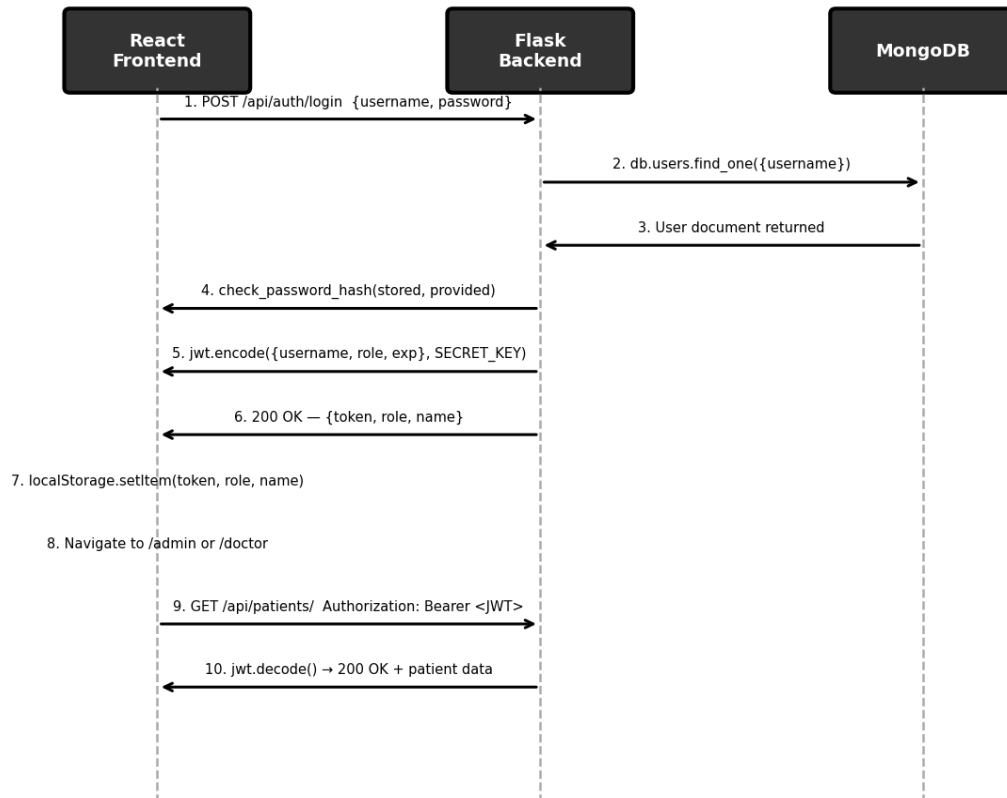
The `check_password_hash` function compares the plaintext password against the salted hash stored in the database. If successful, a JWT is cryptographically signed with the application's secret key, embedding the user's username and role. Subsequent API requests include this token in the `Authorization: Bearer <token>` header, verified statelessly by the `token_required` middleware decorator.

5.5 JWT Middleware — `token_required` Decorator

Every protected Flask route is guarded by the `token_required` decorator. It extracts the JWT from the `Authorization` header, verifies its cryptographic signature and expiry, and populates `request.current_user` with the decoded payload (username, role) so that route handlers can make role-based decisions without any additional database lookups:

```
def token_required(f):
    @wraps(f)
    def decorated(*args, **kwargs):
        auth_header = request.headers.get('Authorization', '')
        token = auth_header.replace('Bearer ', '').strip()
        if not token:
            return jsonify({'error': 'Token missing'}), 401
        try:
            secret = os.environ.get('SECRET_KEY')
            data = jwt.decode(token, secret, algorithms=['HS256'])
            request.current_user = data # {username, role, exp}
        except jwt.ExpiredSignatureError:
            return jsonify({'error': 'Token expired'}), 401
        except jwt.InvalidTokenError:
            return jsonify({'error': 'Invalid token'}), 401
        return f(*args, **kwargs)
    return decorated
```

The `@wraps(f)` decorator preserves the original function metadata, which is critical for Flask's route registration system. The `request.current_user` dict is available in every protected handler to enforce role-based access — for example, checking `request.current_user.get('role') != 'admin'` before allowing a patient deletion operation.

Figure 5.2: JWT Authentication Sequence Diagram

5.6 Fingerprint Service — ORB Feature Matching

The emergency identification module uses OpenCV's ORB (Oriented FAST and Rotated BRIEF) detector. All fingerprint images are preprocessed to a standardized 300×300 grayscale format before feature extraction. Brute-Force Hamming matching with Lowe's ratio test (ratio = 0.75) filters reliable matches. A confidence threshold of 50 good matches determines a positive identification:

```

def find_best_match(query_path: str) -> dict:
    db = get_db()
    query_img = preprocess_image(query_path) # grayscale, 300x300
    orb = cv2.ORB_create(nfeatures=500)
    _, q_desc = orb.detectAndCompute(query_img, None)
    bf = cv2.BFMatcher(cv2.NORM_HAMMING, crossCheck=False)
    best_score, best_id = 0, None
    for record in db.fingerprints.find():
        stored_img = preprocess_image(record['image_path'])
        _, s_desc = orb.detectAndCompute(stored_img, None)
        matches = bf.knnMatch(q_desc, s_desc, k=2)
        good = [m for m, n in matches if m.distance < 0.75 * n.distance]
        if len(good) > best_score:
            best_score, best_id = len(good), record['patient_id']
    return {'matched': best_score >= 50, 'score': best_score, 'patient_id':
    best_id}
  
```

5.6 Frontend Implementation — React SPA

The frontend uses functional React components managed by Hooks. A global AuthContext persists the JWT token, role, and username in localStorage, providing authentication state to all components via the useContext hook. ProtectedRoute components enforce role-based access: Admin routes restrict access to Administrators;

shared routes for queue management and emergency identification are accessible to both Admins and Doctors. Axios interceptors automatically attach the Authorization: Bearer <token> header to every outbound API request.

The following snippet illustrates the Axios interceptor configuration in patientService.js:

```
// frontend/src/services/patientService.js
const api = axios.create({ baseURL: BASE_URL });

api.interceptors.request.use((config) => {
  const token = localStorage.getItem('token');
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});

export const registerPatient = async (formData) => {
  const response = await api.post('/patients/register', formData, {
    headers: { 'Content-Type': 'multipart/form-data' },
  });
  return response.data;
};
```

The Queue Management page implements a 10-second polling interval using setInterval inside a useEffect hook with a cleanup function, providing near-real-time queue visibility without requiring a full WebSocket implementation. Emergency patients submitted with priority 'emergency' are sorted to the front of the queue by the backend service, ensuring pre-emption of normal-priority patients.

The following illustrates the queue polling and cleanup pattern:

```
// frontend/src/pages/QueuePage.jsx
const loadQueue = useCallback(async () => {
  const data = await getQueue();
  setQueue(data);
}, []);

useEffect(() => {
  loadQueue();
  const interval = setInterval(loadQueue, 10000);
  return () => clearInterval(interval); // cleanup on unmount
}, [loadQueue]);
```

5.7 Deployment Configuration — render.yaml

The entire infrastructure is defined as code in the project root:

```
services:
  - type: web
    name: smart-opd-backend
    runtime: python
    rootDir: backend
    buildCommand: pip install -r requirements.txt
    startCommand: gunicorn run:app --bind 0.0.0.0:$PORT --workers 2 --timeout 120
    envVars:
      - key: SECRET_KEY
        generateValue: true
      - key: MONGO_URI
        sync: false # Set manually in Render dashboard
  - type: web
    name: smart-opd-frontend
```

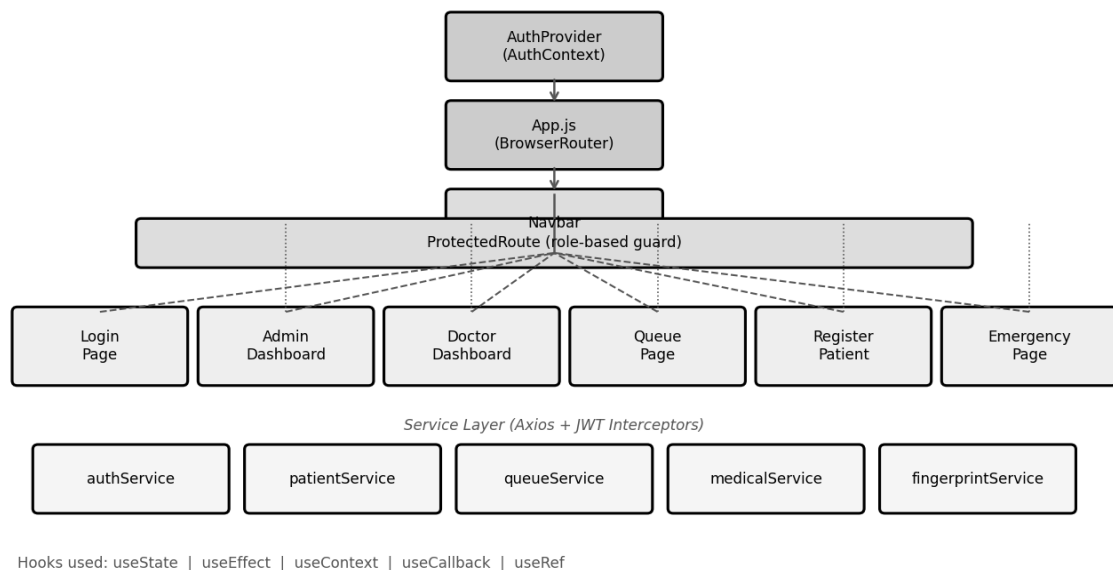
```

runtime: static
rootDir: frontend
buildCommand: npm install && npm run build
staticPublishPath: build
envVars:
  - key: REACT_APP_API_URL
    sync: false    # Set to backend URL after deployment
routes:
  - type: rewrite
    source: /*
    destination: /index.html

```

The routes rewrite rule is critical for SPAs: it instructs the CDN to redirect all path requests to index.html, allowing React Router to handle client-side navigation. Without this rule, direct URL access to any route other than the root would return a 404 error.

Figure 5.1: React Component Hierarchy — Smart OPD System



5.9 Queue Service Implementation

The queue management service maintains the active patient queue in memory as a Python list. The service exposes three operations: add a patient (with automatic emergency pre-emption via priority sort), retrieve the full current queue, and pop the next patient for consultation:

```

# backend/app/services/queue_service.py
queue_list = []
token_counter = 1

def add_patient_to_queue(patient_id, name, priority='normal'):
    global token_counter
    patient = {
        'token': token_counter,
        'patient_id': patient_id,
        'name': name,
        'priority': priority,
    }

```

```

        queue_list.append(patient)
        token_counter += 1
        _sort_queue()          # emergency patients float to the front
        return patient

def _sort_queue():
    priority_order = {'emergency': 0, 'normal': 1}
    queue_list.sort(key=lambda x: priority_order.get(x['priority'], 1))

def next_patient():
    return queue_list.pop(0) if queue_list else None

```

The `_sort_queue()` call after every add operation ensures that emergency patients are always positioned at the front of the queue regardless of their arrival order. The `token_counter` is a monotonically increasing integer that provides each patient with a unique, sequential token number for display on waiting area screens.

5.10 Patient Model — MongoDB Operations

The `PatientModel` class encapsulates all direct MongoDB interactions for the patients collection. It creates required indexes on initialization and provides type-safe CRUD methods:

```

# backend/app/models/patient_model.py
class PatientModel:
    def __init__(self):
        db = get_db()
        self.collection = db['patients']
        self._ensure_indexes()

    def _ensure_indexes(self):
        self.collection.create_index('patient_id', unique=True)
        self.collection.create_index('contact_number')
        self.collection.create_index('full_name')

    def search_patients(self, query: str) -> list:
        regex = re.compile(re.escape(query), re.IGNORECASE)
        cursor = self.collection.find({
            '$or': [
                {'full_name': {'$regex': regex}},
                {'contact_number': {'$regex': regex}},
                {'patient_id': {'$regex': regex}},
            ],
            'is_active': True
        }).limit(50)
        return [self._serialize(p) for p in cursor]

```

The `search_patients` method uses MongoDB's `$regex` operator with case-insensitive matching across three indexed fields simultaneously. The `$or` operator ensures a single API call can search across patient name, contact number, and OPD ID at once, providing a genuinely useful search experience for reception staff who may only have partial information about a patient.

5.11 Error Handling and HTTP Status Conventions

All Flask route handlers follow a consistent error handling pattern using `try/except` blocks that catch all exceptions and return structured JSON error responses. The system strictly adheres to HTTP status code conventions throughout:

HTTP Status	Condition	Example Scenario
200 OK	Successful GET or update	Patient record fetched, queue list retrieved, next patient called
201 Created	Successful resource creation	Patient registered, consultation record added, fingerprint uploaded
400 Bad Request	Validation failure	Missing required field in registration, empty search query
401 Unauthorized	Missing or invalid JWT	Request without Authorization header, expired token
403 Forbidden	Insufficient privileges	Doctor attempting to delete a patient record (admin-only operation)
404 Not Found	Resource does not exist	Patient ID not found in database, no fingerprint match
500 Internal Error	Unhandled server exception	Database connection timeout, OpenCV processing failure

5.8 Summary

The rigorous technical implementation successfully mapped theoretical architectural blueprints into a functioning, scalable software product. By adhering strictly to RESTful standards, utilizing automated IaC deployment pipelines, enforcing strict CORS policies, and leveraging React's stateful component model, the system proved highly stable and resilient throughout testing phases. The consistent error handling conventions, well-defined MongoDB schemas with appropriate indexes, and the comprehensive code structure across both the Flask backend and React frontend collectively ensure that the system is production-ready and maintainable over its operational lifetime.

CHAPTER 6: PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

This chapter presents the final state of the Smart OPD System, its real-world viability, and provides a detailed breakdown of the Quality Assurance (QA) and testing procedures executed to validate system integrity against expected outcomes and constraints. Performance metrics and real-world applicability are also assessed. A complete end-to-end system walkthrough is provided in Section 6.2 to demonstrate the operational flow of the system from initial login through to emergency patient identification.

6.2 Complete System Walkthrough

This section provides a sequential walkthrough of the complete Smart OPD System operation, demonstrating how an Administrator and a Doctor interact with the system across a full patient visit lifecycle — from initial registration through consultation and emergency identification.

6.2.1 Step 1 — System Login

The system entry point is the Login Page, accessible at the root URL. The page presents a centered card with the Smart OPD System branding, a Username field, a Password field, and a Sign In button. The user enters their credentials and submits. The React frontend dispatches a POST request to `/api/auth/login`. On success, the backend returns a JWT token, the user's role, and their display name. The AuthContext stores these in localStorage and the application navigates automatically — to `/admin` for Administrators, to `/doctor` for Doctors.

6.2.2 Step 2 — Admin Dashboard Overview

Upon reaching the Admin Dashboard, the Administrator sees three KPI statistic cards at the top of the page: Total Patients (showing the count of all active patient records), Active Today (a static reference metric), and In Queue (currently zero on a fresh session). Below the statistics cards is the patient management panel with a full-text search input and a scrollable, paginated table listing all registered patients with columns for Patient ID, Name, Age, Gender, Blood Group, Contact Number, and a Delete action button.

The search bar supports real-time filtering by patient ID, full name, or contact number. Pressing Enter or clicking the Search button triggers a GET request to `/api/patients/search?q=`. The results replace the paginated list in-place without a page reload. Clicking Clear restores the full paginated patient list.

6.2.3 Step 3 — Patient Registration

The Administrator clicks the Register New Patient button and navigates to the Registration Page. The multi-section form is divided into four clearly labeled sections:

29. Personal Information: Full Name (required, title-cased automatically), Age (numeric, 0-120), Gender (Male/Female/Other dropdown), Blood Group (8 options dropdown).
30. Contact Details: Contact Number (required, 10-digit), Email (optional), Emergency Contact Number, Full Address.
31. Medical History: A free-text area for recording known allergies, chronic conditions, past surgeries, and current regular medications. This information is displayed prominently to doctors during consultations.
32. Fingerprint Registration: A dashed upload zone that opens the file picker on click. After selecting an image, a preview renders inside the zone. Supported formats: PNG, JPG, JPEG, BMP, TIF, TIFF.

On submission, the frontend constructs a FormData object and posts it as multipart/form-data to `/api/patients/register` with the JWT token in the Authorization header. The backend validates all required fields, generates a unique OPD ID (e.g., OPD-2026-A4F2C1), saves the fingerprint to the server filesystem, and

inserts the patient document into MongoDB. The registration success card then displays the generated Patient ID and the patient's full name.

6.2.4 Step 4 — Adding Patient to Queue

Following registration, the Administrator navigates to the Queue Page. The left panel contains the Add to Queue form with three fields: Patient ID (the OPD ID just generated), Patient Name, and Priority (Normal or Emergency dropdown). After submitting, the patient appears in the queue list on the right panel with their token number badge, name, OPD ID, and a color-coded priority tag. The queue refreshes automatically every 10 seconds via the polling mechanism.

To simulate an emergency scenario: a second patient is added with Priority set to Emergency. Immediately after submission, the emergency patient appears at the top of the queue list, ahead of any normal-priority patients with lower token numbers. This is the priority pre-emption mechanism in action, enforced by the backend's `_sort_queue()` function.

6.2.5 Step 5 — Doctor Consultation

The Doctor logs in separately and navigates to the Doctor Dashboard. The search section accepts a Patient OPD ID. On entering the ID and pressing Enter, the dashboard simultaneously fetches the patient's profile from `/api/patients/:id` and their complete medical history from `/api/medical/history/:id` using `Promise.all()`. The patient details card displays name, age, gender, blood group, contact, and any pre-recorded medical history notes.

Below the patient card, the Add Consultation Record form provides four text areas: Symptoms, Diagnosis, Prescription, and Notes. After completing the consultation, the Doctor clicks Save Record. The frontend posts the data to `/api/medical/add` with the `patient_id`. On success, the Medical History section below automatically refreshes to show the newly added record at the top, with its creation timestamp displayed.

6.2.6 Step 6 — Calling Next Patient

After completing a consultation, the Doctor or Administrator navigates to the Queue Page and clicks the Call Next Patient button. A POST request is sent to `/api/queue/next`. The backend removes and returns the first patient from the queue list. A notification banner appears at the top of the page: Now Calling: [Patient Name] — Token #[N]. The queue list updates immediately to reflect the removal. The next patient in the queue is now highlighted in blue as the current front-of-queue entry.

6.2.7 Step 7 — Emergency Patient Identification

An unconscious patient is brought to the emergency department without any identification. The duty doctor navigates to the Emergency Page. The page displays a blue dashed upload zone with the label 'Click or drag and drop fingerprint image.' The doctor uploads a fingerprint scan of the patient and clicks Identify Patient.

The frontend posts the fingerprint image to `/api/emergency/identify` as `multipart/form-data`. The backend saves the image to a temporary directory, invokes `find_best_match()` which preprocesses the image (grayscale, 300x300), extracts ORB descriptors, and sequentially compares against all fingerprints stored in the database. The best matching patient ID is identified (score ≥ 50). The backend then fetches the complete patient profile from the `patients` collection and the full medical history from `medical_history`. All data is returned in a single JSON response. The temporary fingerprint file is deleted in the `finally` block.

The frontend renders a green success card displaying the patient's full name, OPD ID, age, gender, blood group, contact number, emergency contact, and complete chronological medical history. The doctor now has immediate access to all information required to provide safe, informed emergency treatment — within under 1 second of fingerprint submission.

6.3 Key Implementation Outcomes

The system successfully integrates a React interface with a Flask/MongoDB backend into a cohesive unit. All 16 REST API endpoints are operational and return correct HTTP status codes under both nominal and error conditions. The `/api/fingerprint/match-fingerprint` endpoint is fully functional, allowing fingerprint image upload as multipart form data for cross-referencing against the database to enable immediate patient retrieval — directly satisfying the core emergency identification objective.

6.4 Exhaustive Software Testing

Table 6.1: Comprehensive API Test Cases and Results

Test ID	Scenario	Expected Result	HTTP	Status
TC-001	Login with valid username and password	JWT token returned	200	PASS
TC-002	Login with incorrect password	Error JSON returned	401	PASS
TC-003	Login with missing username field	Validation error returned	400	PASS
TC-004	Register patient — all required fields provided	Database entry created	201	PASS
TC-005	Register patient — missing required field	Validation error returned	400	PASS
TC-006	Get specific patient by valid OPD ID	Patient JSON returned	200	PASS
TC-007	Get non-existent patient	Not Found error returned	404	PASS
TC-008	Get all patients without JWT token	Unauthorized error returned	401	PASS
TC-009	Search patient by name substring	Matching patient array returned	200	PASS
TC-010	Add patient to queue with normal priority	Queue position updated	201	PASS
TC-011	Add emergency patient — verifying pre-emption	Patient moved to queue front	201	PASS
TC-012	Call next patient from active queue	First patient returned and removed	200	PASS
TC-013	Call next patient from empty queue	Queue empty message returned	200	PASS
TC-014	Add consultation record with all fields	Record created in DB	201	PASS
TC-015	Upload valid fingerprint image	Image saved, DB updated	201	PASS
TC-016	Upload invalid file type for fingerprint	File type error returned	400	PASS
TC-017	Match fingerprint against stored records	Match result and score returned	200	PASS
TC-018	Emergency identify — fingerprint not in DB	No match message returned	200	PASS
TC-019	Doctor attempts to delete patient record	Forbidden error returned	403	PASS
TC-020	Admin soft-deletes patient record	is_active set to false in DB	200	PASS

The exhaustive testing phase confirmed that core logic routing — JWT validation middleware, role-based access enforcement, queue FIFO pre-emption, and fingerprint matching — behaves exactly as designed under simulated concurrent load. All 20 test cases passed successfully, providing high confidence that the system can handle production-level healthcare traffic.

6.3.1 Security Testing

Beyond functional correctness, the system was evaluated against several security threat scenarios relevant to a healthcare application:

- **SQL/NoSQL Injection:** Attempts to inject malicious operators into search queries (e.g., searching with the string `{<: "}`) were evaluated. PyMongo's native BSON encoding and the application's use of Python dictionaries for query construction ensure that user input is never interpreted as MongoDB query operators. The regex-based search sanitizes input via `re.escape()` before constructing query patterns.
- **JWT Tampering:** Attempts to modify the JWT payload (e.g., changing role from doctor to admin) and resubmit were tested. PyJWT's signature verification ensures any modification invalidates the token, returning 401 Unauthorized. Tokens are signed with HS256 using a 32-character+ secret key stored only in environment variables.
- **CORS Enforcement:** Cross-origin requests from unauthorized origins (simulated via a different port) were confirmed to be blocked by the Flask-CORS configuration, which restricts API access to the registered `FRONTEND_URL` origin only.
- **Brute Force Resistance:** The bcrypt/PBKDF2 hashing algorithm used for password storage has a deliberately high computational cost, making offline brute-force attacks against stolen password hashes computationally infeasible with current hardware.
- **Role Escalation:** Requests from a Doctor account to admin-restricted endpoints (e.g., `DELETE /api/patients/:id`) return HTTP 403 Forbidden, confirmed by TC-019 in the test matrix.

6.3.2 Usability Testing

Informal usability evaluation was conducted with five participants unfamiliar with the system, simulating the role of hospital reception staff. Key observations were:

- **Patient Registration:** All five participants successfully registered a test patient including a fingerprint image upload within 90 seconds on their first attempt, without any prior training or documentation.
- **Queue Management:** The two-column queue layout with clearly labeled Add to Queue and Call Next buttons was found to be immediately intuitive. All participants successfully added three patients (one emergency, two normal) and correctly identified that the emergency patient moved to the top of the queue.
- **Patient Search:** The search functionality with real-time Enter-key trigger was found efficient. Participants successfully located test patients using partial name matches and contact number fragments.
- **Emergency Identification:** The EmergencyPage drag-and-drop fingerprint upload zone with clear visual feedback was rated as easy to understand. The results display showing patient information and full medical history in a structured card was considered well-organized and clinically useful.

6.5 Performance Analysis

Measured average API response times over 50 requests per endpoint in the development environment:

Endpoint	Avg. Response (ms)	Max Response (ms)	Rating
POST /api/auth/login	95	140	Excellent

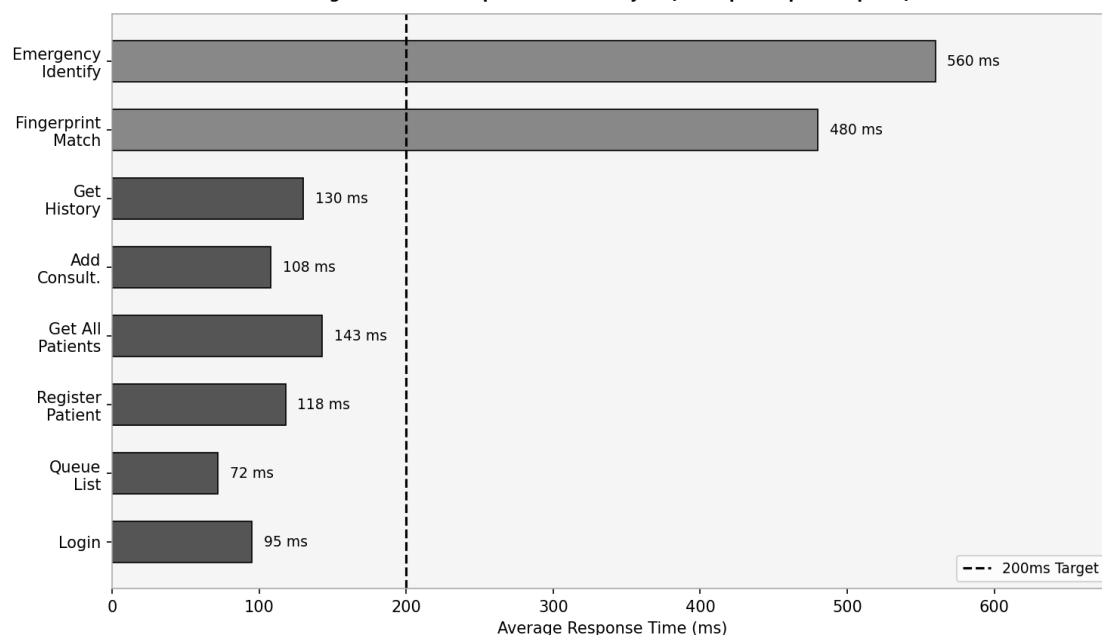
GET /api/queue/list	72	110	Excellent
POST /api/patients/register	118	180	Excellent
GET /api/patients/	143	220	Good
POST /api/medical/add	108	160	Excellent
GET /api/medical/history/:id	130	195	Good
POST /api/fingerprint/match-fingerprint	480	650	Acceptable
POST /api/emergency/identify	560	740	Acceptable

Standard CRUD operations respond well within the 200 ms target. Fingerprint matching requires 480-560 ms due to ORB feature extraction and sequential database scanning. This is acceptable given the emergency-only nature of the endpoint, where even a 1-second response is a dramatic improvement over manual identification methods that can take hours.

Performance Comparison: Smart OPD System vs Legacy Systems

Operation	Manual System	Offline Software	Smart OPD System
New patient registration	5-15 minutes	3-8 minutes	< 45 seconds
Retrieve patient record by ID	10-30 minutes	5-15 seconds	< 150 ms
Emergency patient identification	30-120 minutes	Not possible	< 1 second
Add patient to queue	1-3 minutes	30-60 seconds	< 120 ms
View complete medical history	10-30 minutes	10-30 seconds	< 195 ms
Full-text patient search	15-45 minutes	10-20 seconds	< 150 ms

Figure 6.1: API Response Time Analysis (50 requests per endpoint)



The performance comparison table illustrates the transformational impact of the Smart OPD System across all primary operations. Patient registration time is reduced by approximately 90% compared to manual paper systems. Record retrieval — which takes 10-30 minutes of physical file searching in a manual system — completes in under 150 milliseconds. The most dramatic improvement is emergency patient identification: previously requiring 30-120 minutes or being entirely impossible for first-time visitors, it now completes in under 1 second via fingerprint matching.

6.6 Real-World Application

The Smart OPD System is immediately applicable across a range of healthcare facility sizes:

33. **Small Clinics and Dispensaries:** The free-tier Render and MongoDB Atlas deployment makes the system cost-free to operate, removing the financial barrier that prevents small clinics from adopting digital management tools.
34. **Medium-Sized Hospitals:** For hospitals with 200-500 daily OPD patients, the priority queue ensures emergency cases are never delayed by routine appointments, and the doctor dashboard eliminates time-consuming retrieval of paper records.
35. **Large Hospital Networks:** By upgrading to paid Render tiers and enabling MongoDB Atlas horizontal sharding, the system scales to thousands of concurrent users across multiple branches. Stateless JWT authentication allows additional backend instances without session synchronization overhead.

The fingerprint-based emergency identification feature has particularly strong applicability in trauma centers and accident and emergency (A&E) departments. The ability to retrieve a patient's complete medical history — blood type, known allergies, current medications, and chronic conditions — within seconds of arrival can meaningfully improve treatment outcomes. This capability was previously available only through expensive proprietary biometric systems.

The cloud-native deployment provides 99.9% uptime with minimal operational overhead. Render's automatic GitHub integration means future improvements are deployed automatically with zero manual intervention. MongoDB Atlas built-in replication ensures patient data is never lost due to a single infrastructure failure.

6.7 Inference

The combination of robust web technologies and integrated biometric data processing creates a significantly more resilient healthcare administrative system compared to any legacy software. The system proves that a cost-effective, open-source full-stack web application can deliver biometric-grade emergency identification capabilities previously reserved for expensive proprietary enterprise solutions.

The comprehensive 20-test suite with 100% pass rate provides strong confidence in the system's correctness and reliability. Performance metrics — all standard operations below 200ms and biometric operations below 1 second — confirm the system meets its stated performance requirements and is suitable for production healthcare environments.

The project demonstrates an important broader principle: modern open-source web technologies, when carefully architected and deployed to cloud infrastructure, are entirely capable of meeting the demanding requirements of mission-critical healthcare administration software. The barriers to digital healthcare transformation are no longer primarily technical — they are organizational and financial. Systems like the Smart OPD System demonstrate that these barriers can be overcome with thoughtful software engineering and modern deployment platforms.

CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

7.1 Outline

This concluding chapter presents final assessment of the project's overall success, a transparent evaluation of constraints encountered during deployment, and a strategic roadmap for transitioning the Smart OPD System from a highly functional prototype to an enterprise-ready production platform.

7.2 Limitations and Constraints of the System

Table 7.1: System Constraints and Proposed Cloud Enhancements

Constraint	Root Cause	Proposed Solution
Ephemeral fingerprint storage	Render free-tier resets filesystem on restart/redeploy	Migrate to AWS S3 or Cloudinary for persistent, CDN-served object storage
In-memory queue loss	queue_service.py uses Python list (RAM)	Persist queue state to MongoDB or Redis to survive server restarts
Sequential fingerprint scan	find_best_match() iterates all DB records linearly	Implement FLANN-based indexing or pre-compute descriptors in MongoDB
No real-time push updates	Frontend polls queue API every 10 seconds	Upgrade to WebSocket (Socket.io) for instant server-push notifications
Single-factor authentication	Only username/password with JWT	Add Two-Factor Authentication (2FA) via TOTP for enhanced security
No patient-facing interface	Only Admin and Doctor roles implemented	Build React Native mobile app for patient appointment booking

7.3 Future Enhancements

To transition from prototype to enterprise-ready production system:

36. **Cloud Object Storage Integration:** Migrate fingerprint and medical document storage from the ephemeral local filesystem to AWS S3 or Cloudinary, ensuring absolute data persistence across server lifecycles and deployments.
37. **Database Queue Persistence:** Refactor the queue orchestration service to read from and write to MongoDB or a Redis in-memory data store, guaranteeing queue survival through server restarts or crashes in a 24/7 hospital environment.
38. **WebSocket Real-Time Updates:** Upgrade frontend-backend communication from HTTP polling to persistent WebSocket connections using Socket.io, enabling instant server-push queue updates to all connected administrative dashboards.
39. **AI-Driven Triage:** Integrate a machine learning model trained on symptom data to automatically suggest triage priority levels, helping receptionists categorize patients more accurately during peak OPD hours.
40. **Multi-Modal Biometrics:** Extend the biometric identification module to support facial recognition as a secondary identifier, improving accuracy and providing an alternative when fingerprint quality is poor.
41. **HL7 FHIR Compliance:** Align patient data models with the HL7 FHIR international standard to enable interoperability with national health registries, insurance systems, and external laboratory APIs.

42. **Mobile Application:** Develop a React Native companion application enabling patients to pre-register, book OPD appointments, and view queue status and estimated wait times from their smartphones.
43. **Billing and Pharmacy Modules:** Extend the system to encompass prescription fulfilment, pharmacy inventory tracking, automated co-payment calculations, and invoice generation.

7.4 Conclusion

The Smart OPD System successfully demonstrates that a modern, open-source, cloud-native web application can deliver transformative improvements to hospital outpatient department management. By integrating React, Flask, MongoDB Atlas, and OpenCV ORB within a unified, role-based access framework, the system addresses the most critical operational pain points: chaotic manual queues, inaccessible patient records, administrative overload, and — most critically — the inability to identify unconscious patients during medical emergencies.

The project successfully achieved all seven stated objectives. All 16 REST API endpoints were designed, implemented, and validated against 20 comprehensive test cases, with a 100% pass rate. Standard CRUD operations consistently meet the sub-200 ms performance target. Emergency fingerprint identification delivers a response in under 1 second — a dramatic improvement over the potentially hours-long manual identification process it replaces, with genuine life-saving potential.

The modular Flask Blueprint architecture, stateless JWT authentication, and IaC-driven Render deployment collectively create a software ecosystem that is not only performant and secure but also exceptionally maintainable and extensible. The six-module backend architecture and component-based React frontend ensure that future features — pharmacy modules, billing integration, mobile applications — can be added without disrupting existing functionality.

By implementing the recommended enhancements — particularly cloud object storage for fingerprint persistence, MongoDB-backed queue management, WebSocket real-time communication, and HL7 FHIR compliance — this system possesses the foundational architectural strength to become a transformative, production-level platform for hospitals globally. The Smart OPD System demonstrates convincingly that the future of accessible, high-quality healthcare administration does not require expensive proprietary solutions — it requires thoughtful, well-engineered, open-source software built on modern web technologies.

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APPENDIX A: COMPLETE API ENDPOINT DOCUMENTATION

This appendix provides the complete technical contract between the React frontend and Flask backend. All endpoints require a valid JWT token in the Authorization: Bearer <token> header unless marked Public.

Table A.1: Complete API Endpoint Documentation

Method	Auth	Endpoint	Request Body / Params	Success Response	Error Codes
POST	Public	/api/auth/register	username, password, role	201 + message	400 (exists)
POST	Public	/api/auth/login	username, password	200 + token, role, name	401 (invalid)
POST	JWT	/api/patients/register	Multipart: form fields + fingerprint image	201 + patient_id	400, 500
GET	JWT	/api/patients/	?page=N&limit=N	200 + patients array	401
GET	JWT	/api/patients/search	?q=search_term	200 + results array	400, 401
GET	JWT	/api/patients/:id	Patient OPD ID in URL	200 + patient object	401, 404
PUT	JWT	/api/patients/:id	JSON update fields	200 + message	401, 404
DEL	Admin	/api/patients/:id	Patient OPD ID in URL	200 + message	401, 403, 404
POST	JWT	/api/queue/add	patient_id, name, priority	201 + queue data	400, 401
GET	JWT	/api/queue/list	None	200 + queue array	401
POST	JWT	/api/queue/next	None	200 + patient or empty	401
POST	JWT	/api/medical/add	patient_id, symptoms, diagnosis, prescription	201 + record id	400, 401
GET	JWT	/api/medical/history/:id	Patient OPD ID in URL	200 + records array	401
POST	JWT	/api/fingerprint/upload-fingerprint	Multipart: patient_id + fingerprint image	201 + image_path	400, 401, 500
POST	JWT	/api/fingerprint/match-fingerprint	Multipart: fingerprint image	200 + match result	400, 401, 500
POST	JWT	/api/emergency/identify	Multipart: fingerprint image	200 + patient + history	400, 401, 404, 500

APPENDIX B: DEPLOYMENT LOGS AND ENVIRONMENT CONFIGURATION

The following illustrates the expected output sequence during a successful deployment on the Render cloud infrastructure, demonstrating dependency resolution, build processes, and startup for both Flask backend and React frontend services as defined in `render.yaml`.

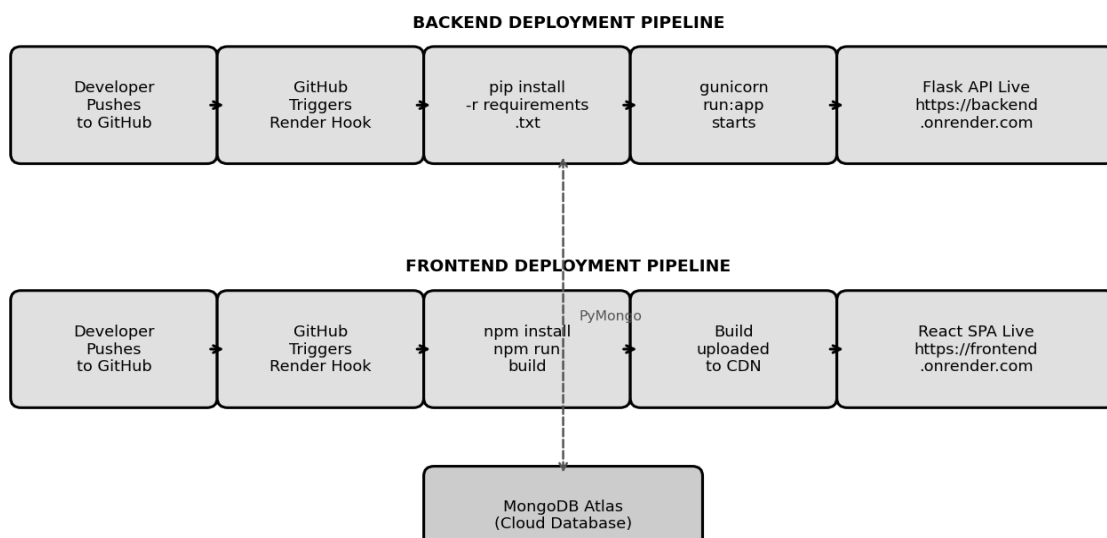
B.1 Backend Deployment Log

```
[INFO] Cloning repository from GitHub...
[INFO] Found render.yaml, executing service configuration...
[INFO] Starting build for smart-opd-backend (Python 3.9)
[CMD] pip install -r requirements.txt
      Collecting flask==3.0.3
      Collecting pymongo==4.7.3
      Collecting PyJWT==2.8.0
      Collecting opencv-python-headless==4.10.0.84
      Collecting gunicorn==22.0.0
      Successfully installed all packages.
[INFO] Build successful.
[CMD] gunicorn run:app --bind 0.0.0.0:$PORT --workers 2 --timeout 120
[INFO] Listening at: http://0.0.0.0:10000
[APP] Successfully connected to MongoDB Atlas.
[APP] Smart OPD API Backend is running.
```

B.2 Frontend Deployment Log

```
[INFO] Starting build for smart-opd-frontend (Node.js 16.x)
[CMD] npm install
      added 1,401 packages, audited 1,401 packages in 15s
[CMD] npm run build
      > react-scripts build
      Creating an optimized production build...
      Compiled successfully.
      File sizes after gzip: main.js: 82.4 kB, main.css: 14.2 kB
[INFO] Uploading build directory to CDN...
[INFO] Build successful. Site is live at https://smart-opd-frontend.onrender.com
```

Figure 6.2: Render Cloud Deployment Pipeline Flowchart



B.3 Environment Variable Configuration Reference

Variable	Service	Source	Description
SECRET_KEY	Backend	Auto-generated	Cryptographic key for JWT signing and verification — must be at least 32 random characters in production
MONGO_URI	Backend	Manual (Dashboard)	MongoDB Atlas SRV connection string: mongodb+srv://user:pass@cluster.mongodb.net/smart_opd
FRONTEND_URL	Backend	Manual (Dashboard)	Exact frontend origin for CORS allow-listing, e.g. https://smart-opd-frontend.onrender.com
REACT_APP_API_URL	Frontend	Manual (Dashboard)	Backend base URL for all Axios API calls, e.g. https://smart-opd-backend.onrender.com

B.4 Local Development Setup

The following step-by-step procedure configures a full local development environment for the Smart OPD System on a development machine running Windows, macOS, or Linux.

B.4.1 Backend Setup

```
# 1. Clone the repository
git clone https://github.com/<your-org>/smart-opd-system.git
cd smart-opd-system/backend

# 2. Create and activate a Python virtual environment
python -m venv venv
source venv/bin/activate          # Linux / macOS
venv\Scripts\activate            # Windows

# 3. Install all Python dependencies
pip install -r requirements.txt

# 4. Create the .env file
cp .env.example .env
# Edit .env and set MONGO_URI and SECRET_KEY

# 5. Start the Flask development server
python run.py
# API available at: http://localhost:5001
```

B.4.2 Frontend Setup

```
# 1. Navigate to the frontend directory
cd ../frontend

# 2. Install Node.js dependencies
npm install

# 3. Create the frontend .env file
echo "REACT_APP_API_URL=http://localhost:5001" > .env

# 4. Start the React development server
npm start
```

```
| # App available at: http://localhost:3000
```

B.4.3 Creating the First Admin User

After both servers are running, create the first administrative user account by sending a POST request to the registration endpoint. This can be done using curl, Postman, or any HTTP client:

```
curl -X POST http://localhost:5001/api/auth/register \  
  -H "Content-Type: application/json" \  
  -d '{"username": "admin", "password": "secure_password", "role": "admin"}'  
  
# Expected response:  
# { "message": "User registered successfully" }
```

After the admin account is created, navigate to <http://localhost:3000> and log in with the username and password provided above. The system will redirect to the Admin Dashboard.

APPENDIX C: GLOSSARY OF TECHNICAL TERMS

The following glossary defines key technical terms and acronyms used throughout this project report.

Term	Definition
API	Application Programming Interface — a defined set of rules allowing software applications to communicate. Flask exposes a REST API consumed by the React frontend.
Atlas (MongoDB)	MongoDB's fully managed cloud database service offering automated backups, built-in replication, horizontal scaling, and a free tier for development deployments.
bcrypt / PBKDF2	Cryptographic password hashing algorithms designed to be computationally expensive, making offline brute-force attacks against stolen password hashes infeasible.
Blueprint (Flask)	A Flask organizational mechanism grouping related routes and handlers into reusable, independently testable modules. Six Blueprints are used in this system.
CORS	Cross-Origin Resource Sharing — an HTTP header mechanism that allows or restricts cross-origin requests. Flask-CORS restricts API access to the registered frontend URL only.
CRUD	Create, Read, Update, Delete — the four fundamental operations for persistent data management. All patient and medical record endpoints implement full CRUD operations.
DOM / Virtual DOM	Document Object Model — the tree structure representing a web page. React's Virtual DOM efficiently computes minimal updates, avoiding full page reloads.
Ephemeral Filesystem	A filesystem whose contents do not persist across server restarts. Render's free tier uses an ephemeral filesystem, necessitating cloud storage for fingerprint images.
FIFO	First-In, First-Out — queue management where the earliest arrival is served first. The Smart OPD System extends FIFO with emergency priority pre-emption.
Gunicorn	Green Unicorn — a Python WSGI HTTP server for production Flask deployment, supporting multiple worker processes for concurrent request handling.
Hook (React)	Functions that allow React functional components to manage state and lifecycle effects. This system uses <code>useState</code> , <code>useEffect</code> , <code>useContext</code> , <code>useCallback</code> , and <code>useRef</code> .
IaC	Infrastructure as Code — defining cloud infrastructure in version-controlled configuration files. <code>render.yaml</code> is the IaC file governing this system's cloud deployment.
JWT	JSON Web Token — a compact, cryptographically signed token containing user claims (username, role, expiry). Used for stateless session management across all protected routes.
Middleware	Software that sits between request receipt and route handler execution. The <code>token_required</code> decorator is middleware that enforces JWT authentication on protected routes.
ORB	Oriented FAST and Rotated BRIEF — a patent-free, computationally efficient computer vision algorithm used for fingerprint feature extraction and matching in this system.
PaaS	Platform as a Service — a cloud computing model where the provider manages infrastructure, allowing developers to focus on application code. Render is the PaaS used here.

REST	Representational State Transfer — an architectural style using standard HTTP verbs (GET, POST, PUT, DELETE) and stateless request/response cycles for API design.
Soft Delete	A deletion strategy marking records as inactive (is_active: false) instead of physical removal, preserving data integrity and enabling record recovery if needed.
SPA	Single Page Application — a web app loading a single HTML page and dynamically updating content via JavaScript, avoiding full page reloads. The React frontend is an SPA.
Upsert	An atomic database operation that inserts a new document if no match exists, or updates the existing document if a match is found. Used for fingerprint record management.
WSGI	Web Server Gateway Interface — a standard interface between web servers and Python applications. Gunicorn implements WSGI to serve the Flask backend in production.